FINANCIAL TRANSACTIONS AND LOANS DASHBOARD

Power BI Project Report

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Tool Used	Microsoft Power BI Desktop

This report explains the dashboard structure, key indicators, visualizations and interactive features.

1. Introduction

The Financial Transactions and Loans Dashboard is a data visualization project created using Microsoft Power BI Desktop. It is designed to analyse financial transactions, customer information, account types and loan status.

The dashboard presents financial information through KPI cards, charts, a transaction summary table and interactive filters. These visuals make it easier to understand transaction activity and explore financial data.

2. Objectives of the Project

• To analyse financial transactions and loan information.

• To display important financial indicators using KPI cards.

• To compare different account types and transaction categories.

• To analyse transaction trends over time.

• To understand the distribution of loans by status.

• To create an interactive dashboard using Power BI.

• To use drill-down, drill-up and tooltip features for detailed analysis.

3. Tools and Power BI Features Used

Microsoft Power BI Desktop: Used to build the report and present the data visually.

KPI cards: Display key figures such as total customers, active accounts, active loans and total transaction value.

Charts and table: Compare account types, loan statuses, transaction values and transaction counts.

Slicers: Filter the report by account type, transaction type and year.

Drill-down and drill-up: Explore transaction activity at year, quarter and month levels.

Tooltip page: Shows additional transaction details when a user points to a data point.

4. Dashboard Overview

The main page is named **FINANCE OVERVIEW**. It contains four KPI cards, an account-type column chart, a transaction-over-time line chart, a loan-status doughnut chart, a transaction-type horizontal bar chart and a transaction summary table.

The report also contains a separate **TRANSACTION TOOLTIP** page. The slicers allow users to focus on selected account types, transaction types or years.

The PBIX report structure shows that the transaction-type bar chart is configured to compare **Total Transaction Value** by transaction type. The summary table separately displays transaction counts and transaction values.

5. Key Performance Indicators (KPIs)

KPI	Value shown in the reference report	What it tells us
Total Customers	1,072	Number of customers represented in the report.
Active Accounts	1,297	Number of active accounts in the dataset.
Active Loans	213	Number of loans currently marked as active.
Total Transaction Value	$122,693,050.81	Combined monetary value of transactions.

These figures are taken from the supplied reference PDF. Please confirm that the same values appear in your Power BI file after opening it and checking that no slicers are applied.

6. Visualizations Used in the Dashboard

6.1 Accounts by Account Type

A column chart compares the number of accounts across Business, Youth, Savings, Payroll and Checking categories. In the reference report, Business accounts have the highest total. This chart helps compare the distribution of accounts.

6.2 Transaction Over Time

A line chart displays transaction activity over time. It can be explored by year, quarter and month. The reference report shows January as the highest month, followed by a noticeable fall in February and smaller changes during the remaining months.

At year level, the reference chart shows relatively high values from 2020 to 2023, followed by a sharp decline in 2024, with lower values in 2025 and 2026. This pattern should be checked against the original dataset to confirm that the date field and measures are working correctly.

6.3 Loans by Loan Status

Loan status	Number of loans	Percentage
Active	213	73.2%
Paid Off	47	16.15%
Overdue	31	10.65%
Total	291	100%

The doughnut chart shows that Active loans form the largest category. Paid Off and Overdue loans make up the remaining portions of the displayed loan portfolio.

6.4 Transactions by Type

The horizontal bar chart compares Deposit, Transfer, Withdrawal and Payment. In the supplied PBIX visual configuration, this chart uses Total Transaction Value as its measure. It therefore compares the monetary value for each transaction type, rather than the number of transactions.

6.5 Transaction Summary

The summary table lists each transaction type with its total transaction count and transaction value. The reference report displays 48,510 total transactions and a total transaction value of $122,693,050.81.

7. Filters and Interactivity

7.1 Account Type Filter

Allows users to select Business, Youth, Savings, Payroll or Checking accounts.

7.2 Transaction Type Filter

Allows users to focus on Deposit, Transfer, Withdrawal or Payment transactions.

7.3 Year Filter

Allows users to select a year and analyse transaction activity for the selected period.

These slicers make the dashboard interactive. The displayed charts and figures can change based on the selected filters.

8. Drill-Down and Drill-Up Analysis

Drill-down and drill-up are Power BI features that help users move between summary-level and detailed-level information. In this project, the Transaction Over Time line chart uses a date hierarchy.

8.1 Year-Level Analysis

The year-level view displays transaction information from 2020 to 2026. The reference report shows higher values in the earlier years and a sharp decline after 2023. This change needs to be validated using the source data.

8.2 Quarter-Level Analysis

The quarter-level view displays Qtr 1, Qtr 2, Qtr 3 and Qtr 4. It helps users compare transaction activity across the four quarters.

8.3 Month-Level Analysis

The month-level view displays January to December. The reference report shows a high value in January, a decline in February and fluctuations across the remaining months.

8.4 How to Use Drill-Down

• Select the Transaction Over Time chart.

• Click the Drill Down icon in the visual header.

• Select a data point to move to the next level of detail.

• Observe how the chart changes.

8.5 How to Use Drill-Up

• Select the Transaction Over Time chart.

• Click the Drill Up icon in the visual header.

• Observe the chart returning to a higher-level view.

• Compare the broader transaction trend.

9. Transaction Tooltip Page

The separate Transaction Tooltip page provides additional information when users hover over a data point in a visual. The PBIX report contains cards for Total Transactions and Total Transaction Value, along with the transaction type name.

For example, when a user points to a Deposit data point, the tooltip can show the Deposit category and its related transaction count and value. This helps users view extra details without leaving the main dashboard.

10. Key Findings

• Business accounts have the highest total among the account types shown in the reference report.

• The reference KPI cards show 1,072 customers and 1,297 active accounts.

• Active loans are the largest loan-status category, with 213 loans.

• The reference loan-status chart contains 291 loans in total.

• The transaction-type visual compares transaction value by category in the supplied PBIX file.

• January shows the highest monthly transaction activity in the reference report.

• The reference summary table shows 48,510 transactions worth $122,693,050.81.

• Drill-down and drill-up allow users to explore transaction activity at different time levels.

11. Benefits of the Dashboard

• **Easy analysis:** Charts and tables present financial information clearly.

• **Quick overview:** KPI cards show important figures in one place.

• **Better comparison:** Visuals help compare account and transaction categories.

• **Trend identification:** The line chart helps identify changes over time.

• **Interactive reporting:** Slicers allow users to filter the information.

• **Detailed exploration:** Drill-down, drill-up and tooltips provide additional detail.

12. Limitations and Data Validation

The accuracy of the dashboard depends on the quality of the source data and the correctness of the measures created in Power BI.

The total transaction count and total transaction value should be checked against the summary table and KPI cards. The sharp decline in the year-level chart after 2023 should also be verified using the original dataset. The figures in this report that came from the reference PDF should be confirmed in the submitted PBIX file with all slicers cleared.

The report describes the visuals and measures identified in the supplied PBIX file. Where exact category-wise values were not available from the PBIX file structure, no additional figures have been invented.

13. Conclusion

The Financial Transactions and Loans Dashboard demonstrates how Microsoft Power BI can turn financial data into meaningful visual information. It provides an overview of customers, accounts, loans and transactions through KPI cards, charts and a summary table.

The slicers, date hierarchy, drill-down, drill-up and tooltip page make the report interactive and allow users to explore the data at different levels. Overall, this project demonstrates the use of Power BI for financial data analysis, reporting and understanding transaction patterns.